1 Shape Gradient

In this section, we derive the shape gradient when the boundary of the domain $\Omega \subset \mathbb{R}^2$ is parameterized by a coefficient vector $\vec{C}$. Following [2, Eqn. (5)], we represent the boundary as

$$\partial\Omega(\vec{C}) = \gamma(\theta; \vec{C}) = r(\theta; \vec{C})\,(\cos(2\pi\theta),\, \sin(2\pi\theta)), \tag{1}$$

where $\gamma(\theta; \vec{C})$ is a $\vec{C}$-dependent map from $[0, 1]$ into $\mathbb{R}^2$, and $r(\theta; \vec{C})$ is a positive, real-valued, 1-periodic radius function.

We model the radius function with a truncated Fourier series,

$$r(\theta; \vec{C}) = c_0 + \sum_{k=1}^{N} \left(a_k \cos(2\pi k\theta) + b_k \sin(2\pi k\theta)\right), \tag{2}$$

and collect the coefficients into the vector

$$\vec{C} = \left[c_0,\, a_1, \ldots, a_N,\, b_1, \ldots, b_N\right]^\top.$$

Lemma 1.1. *Let $\Omega \subset \mathbb{R}^2$ be a domain whose boundary is parameterized by* (1)-(2)*. If we perturb the i-th coefficient in $\vec{C}$, then the perturbed boundary takes the form*

$$\partial\Omega(\vec{C} + \varepsilon e_i) = \{\gamma(\theta; \vec{C} + \varepsilon e_i) \mid \theta \in [0, 1]\}.$$

The boundary mapping satisfies the linear expansion for ε small enough:

$$\gamma(\theta; \vec{C} + \varepsilon e_i) = \gamma(\theta; \vec{C}) + \varepsilon\, \frac{\partial\gamma}{\partial C_i}(\theta; \vec{C}) + o(\varepsilon).$$

More generally, for any perturbation vector $\vec{\alpha}$,

$$\partial\Omega(\vec{C} + \varepsilon\vec{\alpha}) = \partial\Omega(\vec{C}) + \varepsilon \left\{ \sum_{k=0}^{2N} \alpha_k\, \frac{\partial\gamma}{\partial C_k}(\theta; \vec{C}) \;\middle|\; \theta \in [0, 1] \right\} + o(\varepsilon).$$

Proof. Apply the Taylor expansion to

$$r(\theta; \vec{C} + \varepsilon\vec{\alpha}) = r(\theta; \vec{C}) + \varepsilon \sum_k \alpha_k \frac{\partial r}{\partial C_k}(\theta; \vec{C}) + o(\varepsilon),$$

and use the definition

$$\gamma(\theta; \vec{C}) = r(\theta; \vec{C})\big(\cos(2\pi\theta),\, \sin(2\pi\theta)\big).$$

□

We start with the definition of the shape functional found in [1]

Definition 1.1. A shape functional $J(\Omega)$ is said shape differentiable if the mapping

$$W^{1,\infty}(\mathbb{R}^d, \mathbb{R}^d) \longrightarrow \mathbb{R}, \quad \alpha \longmapsto J(\Omega_\alpha)$$

is Fréchet differentiable at $\alpha = 0$, i.e., if there exists a continuous linear form

$$DJ(\Omega) \in W^{1,\infty}(\mathbb{R}^d, \mathbb{R}^d)^*$$

such that the following asymptotics holds:

$$J(\Omega_\alpha) = J(\Omega) + DJ(\Omega)(\alpha) + o(\alpha), \quad \text{where} \quad \frac{|o(\alpha)|}{\|\alpha\|_{W^{1,\infty}(\mathbb{R}^d,\mathbb{R}^d)}} \xrightarrow{\alpha \to 0} 0.$$

The next lemma tells us that for α small enough, the map $I + \alpha$ is a diffeomorphism.

Lemma 1.2. *For any $\alpha \in W^{1,\infty}(\mathbb{R}^d, \mathbb{R}^d)$ such that $\|\alpha\|_{W^{1,\infty}(\mathbb{R}^d,\mathbb{R}^d)} < 1$, the map $I + \alpha$ is a bijection satisfying $(I + \alpha)^{-1} - I \in W^{1,\infty}(\mathbb{R}^d, \mathbb{R}^d)$.*

Proof. Formally, the inverse map is given by

$$(I + \alpha)^{-1} = \sum_{k=0}^{+\infty} (-1)^k \overbrace{\alpha \circ \cdots \circ \alpha}^{k \text{ times}},$$

where the above series is convergent in the norm of $W^{1,\infty}(\mathbb{R}^d, \mathbb{R}^d)$.

□

We now derive the shape gradient of an arbitrary functional.

Proposition 1.1. *Let Ω be a bounded open set of $\mathbb{R}^d$. For any $f \in W^{1,1}(\mathbb{R}^d)$, the functional $J(\Omega)$ defined by*

$$J(\Omega) := \int_\Omega f(x)\, \mathrm{d}x$$

is shape differentiable, and it holds

$$\mathrm{D}J(\Omega)(\alpha) = \int_\Omega \operatorname{div}(f\alpha)\, \mathrm{d}x = \int_\Omega (\nabla f \cdot \alpha + f \operatorname{div}(\alpha))\, \mathrm{d}x, \qquad \alpha \in W^{1,\infty}(\mathbb{R}^d, \mathbb{R}^d).$$

If, in addition, Ω is smooth, then the above formula can be rewritten as

$$\mathrm{D}J(\Omega)(\alpha) = \int_{\partial\Omega} f\, \alpha \cdot n\, \mathrm{d}\sigma$$

where n denotes the outward normal to Ω.

Proof. For a deformation field α, define the perturbed domain

$$\Omega_t := (I + t\alpha)(\Omega), \qquad |t| \text{ small}.$$

By the change of variables formula,

$$J(\Omega_t) = \int_{\Omega_t} f(x)\,dx = \int_{\Omega} f(x + t\alpha(x)) \det(I + t\nabla\alpha(x))\,dx.$$

Using the first-order Taylor expansions

$$f(x + t\alpha(x)) = f(x) + t\,\nabla f(x) \cdot \alpha(x) + o(t),$$

$$\det(I + t\nabla\alpha) = 1 + t\ \operatorname{div}\alpha + o(t),$$

we obtain

$$J(\Omega_t) = \int_{\Omega} \big(f + t(\nabla f \cdot \alpha + f\ \operatorname{div}\alpha)\big)\,dx + o(t).$$

Thus,

$$\mathrm{D}J(\Omega)(\alpha) = \frac{d}{dt} J(\Omega_t)\bigg|_{t=0} = \int_{\Omega} \big(\nabla f \cdot \alpha + f\ \operatorname{div}\alpha\big)\,dx.$$

Since $\operatorname{div}(f\alpha) = \nabla f \cdot \alpha + f\ \operatorname{div}\alpha$, the volume formula follows immediately. If $\partial\Omega$ is smooth, applying the divergence theorem gives

$$\int_{\Omega} \operatorname{div}(f\alpha)\,dx = \int_{\partial\Omega} f\,\alpha \cdot n\,d\sigma,$$

which yields the boundary expression. □

We now work in the setting of Lemma 1.1, where the boundary $\partial\Omega$ is parameterized by (1)-(2). Since $\partial\Omega$ is given by the mapping $\gamma(\theta; \vec{C})$, the boundary form of the shape derivative yields

$$\mathrm{D}J(\Omega)(\alpha) = \int_0^1 f(\gamma(\theta; \vec{C}))\,\alpha(\gamma(\theta; \vec{C})) \cdot n(\gamma(\theta; \vec{C}))\,\|\gamma'(\theta; \vec{C})\|\,\mathrm{d}\theta.$$

Corollary 1.1. *Assuming suitable conditions*

$$\frac{\partial J(\vec{C})}{\partial a_k} = 2\pi \int_0^1 f(\gamma(\theta, \vec{C})) \cos(2\pi k\theta) r(\theta) d\theta$$

$$\frac{\partial J(\vec{C})}{\partial b_k} = 2\pi \int_0^1 f(\gamma(\theta, \vec{C})) \sin(2\pi k\theta) r(\theta) d\theta$$

$$\frac{\partial J(\vec{C})}{\partial c_0} = 2\pi \int_0^1 f(\gamma(\theta, \vec{C})) r(\theta) d\theta$$

Proof. We perturb a single coefficient a_k to $a_k + \eta$ and examine the induced deformation of the domain. From (2), the corresponding perturbation of the radius function is

$$r(\theta; \vec{C} + \eta e_k^a) = r(\theta; \vec{C}) + \eta \cos(2\pi k\theta),$$

where e_k^a denotes the unit vector in the a_k direction. Substituting this into (1), the perturbed boundary satisfies

$$\gamma_\eta(\theta) = \gamma(\theta; \vec{C}) + \eta \, \cos(2\pi k\theta) \, (\cos(2\pi\theta), \sin(2\pi\theta)).$$

Since $\gamma(\theta; \vec{C}) = r(\theta; \vec{C})(\cos(2\pi\theta), \sin(2\pi\theta))$, we may rewrite the perturbation as

$$\gamma_\eta(\theta) = \gamma(\theta; \vec{C}) + \eta \, \frac{\cos(2\pi k\theta)}{r(\theta; \vec{C})} \, \gamma(\theta; \vec{C}).$$

This identifies the associated velocity field α through

$$\gamma_\eta(\theta) = (I + \eta\alpha)\big(\gamma(\theta; \vec{C})\big),$$

from which we obtain

$$\alpha(\gamma(\theta; \vec{C})) = \frac{\cos(2\pi k\theta}{r(\theta; \vec{C})} (\cos(2\pi\theta), \sin(2\pi\theta)) \, .$$

Substituting this expression into the boundary integral and taking the limit $\eta \to 0$, we obtain

$$\begin{aligned}
\frac{\partial J(\vec{C})}{\partial a_k} &= \lim_{\eta \to 0} \frac{J(\Omega_{\vec{C}+\eta e_k^a}) - J(\Omega_{\vec{C}})}{\eta} \\
&= \int_0^1 f(\gamma(\theta; \vec{C})) \, \alpha(\gamma(\theta; \vec{C})) \cdot n(\gamma(\theta; \vec{C})) \, \|\gamma'(\theta; \vec{C})\| \, \mathrm{d}\theta \\
&= \int_0^1 f(\gamma(\theta; \vec{C})) \, \frac{\partial \gamma(\theta; \vec{C})}{\partial a_k} \cdot n(\gamma(\theta; \vec{C})) \, \|\gamma'(\theta; \vec{C})\| \, \mathrm{d}\theta.
\end{aligned}$$

The last equation follows from the fact that

$$\alpha(\gamma(\theta; \vec{C})) = \frac{\partial \gamma(\theta; \vec{C})}{\partial a_k}.$$

We continue with simplifying the dot product between $\frac{\partial \gamma(\theta, \vec{C})}{\partial a_k}$ and $n(\gamma(\theta, \vec{C}))$. To compute the normal vector, we first compute the tangential vector T. If we denote $r_\theta := \frac{\partial r}{\partial \theta}$, then we have

$$T(\theta) = \frac{\gamma'}{||\gamma'||} = \frac{1}{||\gamma'||}(r_\theta \cos(2\pi\theta) - 2\pi r \sin(2\pi\theta), r_\theta \sin(2\pi\theta) + 2\pi r \cos(2\pi\theta)).$$

Since we are in $\mathbb{R}^2$ the normal vector is therefore equal to

$$n = \frac{1}{||\gamma'||}(r_\theta \sin(2\pi\theta) + 2\pi r \cos(2\pi\theta), -r_\theta \cos(2\pi\theta) + 2\pi r \sin(2\pi\theta)).$$

On the other hand, we also have

$$\frac{\partial \gamma(\theta, \vec{C})}{\partial a_k} = \cos(2\pi k\theta)(\cos(2\pi\theta), \sin(2\pi\theta)).$$

Combining everything we get

$$\frac{\partial J(\vec{C})}{\partial a_k} = \int_0^1 f(\gamma(\theta, \vec{C})) \cos(2\pi k\theta) 2\pi r(\theta) \frac{||\gamma'||}{||\gamma'||} d\theta = \int_0^1 f(\gamma(\theta, \vec{C})) \cos(2\pi k\theta) 2\pi r(\theta) d\theta$$

The other two cases are done similarly; we just have to note that

$$\frac{\partial \gamma(\theta, \vec{C})}{\partial b_k} = \sin(2\pi k\theta)(\cos(2\pi\theta), \sin(2\pi\theta))$$

and

$$\frac{\partial \gamma(\theta, \vec{C})}{\partial c_0} = (\cos(2\pi\theta), \sin(2\pi\theta)).$$

□

2 Adjoint equation for the space-dependent Boltzmann equation

In this section, we will derive the adjoint equation for the space-dependent Boltzmann equation with a reflective boundary condition. The goal will be to find the vector

$$\vec{C} = \left[c_0,\, a_1, \ldots, a_N,\, b_1, \ldots, b_N\right]^\top.$$

which minimized the objective function at time $t = T$

$$J_1(\vec{C}) = \int_{\Omega(\vec{C})} \int_{\mathbb{R}^3} g(v, x) f(v, x, T) dv dx.$$

Using Lagrangian multipliers we set the Lagrangian to be

$$\mathcal{J} = \mathcal{J}_1 + \mathcal{J}_2 + \mathcal{J}_3 \tag{3}$$

where

$$\mathcal{J}_1 = \frac{1}{N} \sum_{i=1}^{N} g(v_{M,i}, x_{M,i}) \tag{4}$$

$$\mathcal{J}_2 = \frac{1}{N} \sum_{k=0}^{M-1} \sum_{i=1}^{N} \beta_{k+1,i} \cdot (v_{k+1,i} - \tilde{v}_{ki}) \tag{5}$$

$$\mathcal{J}_3 = \frac{1}{N} \sum_{k=0}^{M-1} \sum_{i=1}^{N} \alpha_{k+1,i} \cdot (x_{k+1,i} - \tilde{x}_{ki}) \tag{6}$$

where both $x'_{k,i}$ and $v'_{k,i}$ split into a couple of different cases

$$\tilde{v}_{k,i} = \begin{cases} v'_{k,i} & \text{if } x'_{k,i} \in \Omega \\ v'_{k,i} - 2\langle \tilde{n}, v'_{k,i} \rangle \tilde{n} & \text{otherwise} \end{cases}$$

$$\tilde{x}_{k,i} = \begin{cases} x'_{k,i} & \text{if } x'_{k,i} \in \Omega \\ x'_{k,i} - 2\left(\langle \tilde{n}, x'_{k,i} \rangle - \tilde{c}\right) \tilde{n} & \text{otherwise} \end{cases},$$

where $x'_{k,i}$ and $v'_{k,i}$ represent the post collision velocity and position and $\tilde{n}$ and $\tilde{c}$ represent the normal vector and constant defining the place at the point where $x'_{k,i} = x_{k,i} + \Delta t v'_{k,i}$ first intersects the boundary $\partial\Omega$. The reflection line is

$$L = \{x : \langle \tilde{n}, x \rangle = \tilde{c}\}.$$

2.1 Computing $\tilde{n}$ and $\tilde{c}$

Recall the outward normal

$$n(\theta, \vec{C}) = \frac{1}{||\gamma'||}(r_\theta \sin(2\pi\theta) + 2\pi r \cos(2\pi\theta), -r_\theta \cos(2\pi\theta) + 2\pi r \sin(2\pi\theta)). \quad (7)$$

where

$$\gamma(\theta; \vec{C}) = r(\theta; \vec{C})\big(\cos(2\pi\theta),\, \sin(2\pi\theta)\big)$$

and

$$r(\theta; \vec{C}) = c_0 + \sum_{k=1}^{N} \left(a_k \cos(2\pi k\theta) + b_k \sin(2\pi k\theta)\right). \quad (8)$$

To find n_{inter} we need to find the angle θ_{inter} where the line segment $\overrightarrow{x_{k,i}x'_{k,i}}$ first intersects the boundary $\partial\Omega$. For simplicity, we will assume that the intersection point is unique.

Lemma 2.1. *Let $x_{k,i} = (x^x_{k,i}, x^y_{k,i})$ have polar coordinates $x_{k,i} = (r_1, \theta_1)$ and let $v'_{k,i} = (v^x_{k,i}, v^y_{k,i})$ be the direction connecting $x_{k,i}$ and $x'_{k,i} = x_{k,i} + \Delta t v'_{k,i}$. If $x'_{k,i}$ has polar coordinates $x'_{k,i} = (r_2, \theta_2)$, then the line segment $\overrightarrow{x_{k,i}x'_{k,i}}$ has the following expression in polar coordinates*

$$r = \frac{v^y_{k,i} x^x_{k,i} - v^x_{k,i} x^y_{k,i}}{v^y_{k,i} \cos(2\pi\theta) - v^x_{k,i} \sin(2\pi\theta)} \quad \theta \in [\theta_1, \theta_2] mod(1) \quad (9)$$

Remark 2.1. The line segment $\overrightarrow{x_{k,i}x'_{k,i}}$ is parametrized starting at θ_1 and ending at θ_2. If $\theta_1 = \theta_2$, then the line containing $\overrightarrow{x_{k,i}x'_{k,i}}$ passes through the origin and the and only depends on the initial $\theta_1 = \theta_2$

Proof. The line segment in Cartesian coordinates is expressed as

$$(x(t), y(t)) = (x^x_{k,i} + t v^x_{k,i}, x^y_{k,i} + t v^y_{k,i}).$$

In other words, we have

$$t = \frac{x - x^x_{k,i}}{v^x_{k,i}} = \frac{y - x^y_{k,i}}{v^y_{k,i}} \implies v^y_{k,i}(x - x^x_{k,i}) = v^x_{k,i}(y - x^y_{k,i}),$$

which implies

$$v^y_{k,i} x - v^x_{k,i} y = v^y_{k,i} x^x_{k,i} - v^x_{k,i} x^y_{k,i}.$$

Substituting $x = r\cos(2\pi\theta)$ and $y = r\sin(2\pi\theta)$

$$\begin{aligned} r(v^y_{k,i}\cos(2\pi\theta) - v^x_{k,i}\sin(2\pi\theta)) =& v^y_{k,i}x^x_{k,i} - v^x_{k,i}x^y_{k,i} \\ r =& \frac{v^y_{k,i}x^x_{k,i} - v^x_{k,i}x^y_{k,i}}{v^y_{k,i}\cos(2\pi\theta) - v^x_{k,i}\sin(2\pi\theta)} \end{aligned}$$

□

To find the angle of intersection θ_{inter}, we thus need to solve

$$r(\theta;\vec{C}) = c_0 + \sum_{k=1}^{N}(a_k\cos(2\pi k\theta) + b_k\sin(2\pi k\theta)) = \frac{v^y_{k,i}x^x_{k,i} - v^x_{k,i}x^y_{k,i}}{v^y_{k,i}\cos(2\pi\theta) - v^x_{k,i}\sin(2\pi\theta)}. \tag{10}$$

Once we have θ_{inter}, we can solve for c_{inter}, which is the constant term in the definition of the normal plane of the boundary at the point $\gamma(\theta_{inter}, \vec{C})$.

$$c_{\text{inter}} = r(\theta_{\text{inter}}, \vec{C})\langle n(\theta_{\text{inter}}, \vec{C}), \begin{bmatrix}\cos(2\pi\theta_{\text{inter}})\\ \sin(2\pi\theta_{\text{inter}})\end{bmatrix}\rangle. \tag{11}$$

2.2 Computing adjoint variables

In this subsection, we compute the adjoint variables β and α. Throughout the section, we will denote any expression depending on θ_{inter} by $\tilde{\theta}$ and omit any dependence on variables not important at that time. For example

$$c_{\text{inter}} = r(\theta_{\text{inter}}, \vec{C})\langle n(\theta_{\text{inter}}, \vec{C}), \begin{bmatrix}\cos(2\pi\theta_{\text{inter}})\\ \sin(2\pi\theta_{\text{inter}})\end{bmatrix}\rangle \to \tilde{c} = \tilde{r}\langle \tilde{n}, \begin{bmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{bmatrix}\rangle \tag{12}$$

By differentiating the Lagrangian (3) at the final time, we immediately get

$$\beta_{M,i} = -\partial_v g(v_{M,i}, x_{M,i}) \tag{13}$$
$$\alpha_{M,i} = -\partial_x g(v_{M,i}, x_{M,i}). \tag{14}$$

The other adjoint variables require more work.

2.2.1 Computing $\frac{\partial(\mathcal{J}_2+\mathcal{J}_3)}{\partial v_{k,i}}$

We first examine the gradient with respect to $v_{k,i}$ where we assume a collision occurs between particles i and i_1

$$\frac{\partial \mathcal{J}_2}{\partial v_{k,i}} = \frac{1}{N}\beta_{k,i} - \frac{1}{N}\left(\frac{\partial \tilde{v}_{ki}}{\partial v_{k,i}}\right)^\top \beta_{k+1,i} - \frac{1}{N}\left(\frac{\partial \tilde{v}_{ki_1}}{\partial v_{k,i}}\right)^\top \beta_{k+1,i_1} \tag{15}$$

and similarly

$$\frac{\partial \mathcal{J}_3}{\partial v_{k,i}} = -\frac{1}{N}\left(\frac{\partial \tilde{x}_{k,i}}{\partial v_{k,i}}\right)^\top \alpha_{k+1,i} - \frac{1}{N}\left(\frac{\partial \tilde{x}_{k,i_1}}{\partial v_{k,i}}\right)^\top \alpha_{k+1,i_1}. \tag{16}$$

Combining the expressions, we get for particle i

$$\begin{aligned}\frac{\partial \mathcal{J}}{\partial v_{k,i}} =& \frac{1}{N}\beta_{k,i} - \frac{1}{N}\left(\frac{\partial \tilde{v}_{ki}}{\partial v_{k,i}}\right)^\top \beta_{k+1,i} - \frac{1}{N}\left(\frac{\partial \tilde{v}_{ki_1}}{\partial v_{k,i}}\right)^\top \beta_{k+1,i_1} \\ & - \frac{1}{N}\left(\frac{\partial \tilde{x}_{k,i}}{\partial v_{k,i}}\right)^\top \alpha_{k+1,i} - \frac{1}{N}\left(\frac{\partial \tilde{x}_{k,i_1}}{\partial v_{k,i}}\right)^\top \alpha_{k+1,i_1}. \end{aligned} \tag{17}$$

and likewise for particle i_1

$$\begin{aligned}\frac{\partial \mathcal{J}}{\partial v_{k,i_1}} =& \frac{1}{N}\beta_{k,i_1} - \frac{1}{N}\left(\frac{\partial \tilde{v}_{ki_1}}{\partial v_{k,i_1}}\right)^\top \beta_{k+1,i_1} - \frac{1}{N}\left(\frac{\partial \tilde{v}_{k,i}}{\partial v_{k,i_1}}\right)^\top \beta_{k+1,i} \\ & - \frac{1}{N}\left(\frac{\partial \tilde{x}_{k,i_1}}{\partial v_{k,i_1}}\right)^\top \alpha_{k+1,i_1} - \frac{1}{N}\left(\frac{\partial \tilde{x}_{k,i}}{\partial v_{k,i_1}}\right)^\top \alpha_{k+1,i}. \end{aligned} \tag{18}$$

The above expressions tell us that we will need to compute four different terms

$$\frac{\partial \tilde{v}_{ki}}{\partial v_{k,i}}, \quad \frac{\partial \tilde{v}_{ki_1}}{\partial v_{k,i}}, \quad \frac{\partial \tilde{x}_{k,i}}{\partial v_{k,i}}, \quad \frac{\partial \tilde{x}_{k,i_1}}{\partial v_{k,i}}.$$

Lemma 2.2.

$$\frac{\partial \tilde{v}_{k,i}}{\partial v_{k,i}} = \begin{cases} J & \textit{if } x'_{k,i} \in \Omega \\ \left[I - 2\langle \tilde{n}, v'_{k,i}\rangle A - 2\tilde{n}\left((v'_{k,i})^\top A + \tilde{n}^\top\right)\right] J & \textit{Otherwise} \end{cases}$$

Where

$$A := \frac{\partial \tilde{n}}{\partial \tilde{\theta}} \frac{\partial \tilde{\theta}}{\partial v'_{k,i}} \qquad J := \frac{\partial v'_{k,i}}{\partial v_{k,i}}.$$

Proof. Using the chain rule, we have

$$\begin{aligned}\frac{\partial}{\partial v_{k,i}}\left(v'_{k,i} - 2\langle \tilde{n}, v'_{k,i}\rangle \tilde{n}\right) =& \frac{\partial v'_{k,i}}{\partial v_{k,i}} - 2\langle \tilde{n}, v'_{k,i}\rangle A \frac{\partial v'_{k,i}}{\partial v_{k,i}} \\ & - 2\tilde{n}\left((v'_{k,i})^\top A + \tilde{n}^\top\right)\frac{\partial v'_{k,i}}{\partial v_{k,i}} \\ =& \left[I - 2\langle \tilde{n}, v'_{k,i}\rangle A - 2\tilde{n}\left((v'_{k,i})^\top A + \tilde{n}^\top\right)\right] J. \end{aligned}$$

If $x'_{k,i} \in \Omega$, then we the intersection does not occur and we have $\tilde{n} = 0$. □

Lemma 2.3.

$$\frac{\partial \tilde{x}_{k,i}}{\partial v_{k,i}} = \begin{cases} \Delta t J & \textit{if } x'_{k,i} \in \Omega \\ \left[\Delta t I - 2\tilde{n}\left((x'_{k,i})^T A + \Delta t \tilde{n}^T - \frac{\partial \tilde{c}}{\partial v'_{k,i}}\right) - 2(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c})A\right] J, & \textit{Otherwise} \end{cases}$$

Proof. As before, we use the chain rule and $x'_{k,i} = x_{k,i} + \Delta t v'_{k,i}$

$$\begin{aligned}
&\frac{\partial}{\partial v_{k,i}} \left(x'_{k,i} - 2(\langle \tilde{n}, x'_{k,i} \rangle - \tilde{c})\tilde{n} \right) \\
=&\Delta t \frac{\partial v'_{k,i}}{\partial v_{k,i}} \\
&- 2\Big[\tilde{n} \left((x'_{k,i})^\top A + \Delta t\, \tilde{n}^\top - \frac{\partial \tilde{c}}{\partial v'_{k,i}} \right) + (\langle \tilde{n}, x'_{k,i} \rangle - \tilde{c})\, A \Big] \frac{\partial v'_{k,i}}{\partial v_{k,i}},
\end{aligned}$$

□

Define

$$J^1 := \frac{\partial v'_{k,i_1}}{\partial v_{k,i}}$$

Lemma 2.4.

$$\frac{\partial \tilde{v}_{k,i_1}}{\partial v_{k,i}} = \begin{cases} J^1 & \textit{if } x'_{k,i_1} \in \Omega \\ \left[I - 2\langle \tilde{n}_1, v'_{k,i_1} \rangle A^1 - 2\tilde{n}_1 \left((v'_{k,i_1})^\top A^1 + \tilde{n}_1^\top \right) \right] J^1 & \textit{Otherwise} \end{cases}$$

Where

$$A^1 := \frac{\partial \tilde{n}_1}{\partial \tilde{\theta}_1} \frac{\partial \tilde{\theta}_1}{\partial v'_{k,i_1}}$$

Proof. Chain rule and similar arguments as before

$$\frac{\partial}{\partial v_{k,i}} \left(v'_{k,i_1} - 2\langle \tilde{n}_1, v'_{k,i_1} \rangle \tilde{n}_1 \right) = \left[I - 2\langle \tilde{n}_1, v'_{k,i_1} \rangle A^1 - 2\tilde{n}_1 \left((v'_{k,i_1})^\top A^1 + \tilde{n}_1^\top \right) \right] J^1.$$

□

Lemma 2.5.

$$\frac{\partial \tilde{x}_{k,i_1}}{\partial v_{k,i}} = \begin{cases} \Delta t J^1 & \textit{if } x'_{k,i_1} \in \Omega \\ \left[\Delta t I - 2\tilde{n}_1 \left((x'_{k,i_1})^\top A^1 + \Delta t\, \tilde{n}_1^\top - \frac{\partial \tilde{c}_1}{\partial v'_{k,i_1}} \right) - 2(\langle \tilde{n}_1, x'_{k,i_1} \rangle - \tilde{c}_1)\, A^1 \right] J^1, & \textit{Otherwise} \end{cases}$$

Proof.

$$\begin{aligned}
&\frac{\partial}{\partial v_{k,i}} \left(x'_{k,i_1} - 2(\langle \tilde{n}_1, x'_{k,i_1} \rangle - \tilde{c}_1)\tilde{n}_1 \right) \\
=&\left[\Delta t I - 2\tilde{n}_1 \left((x'_{k,i_1})^\top A^1 + \Delta t\, \tilde{n}_1^\top - \frac{\partial \tilde{c}_1}{\partial v'_{k,i_1}} \right) - 2(\langle \tilde{n}_1, x'_{k,i_1} \rangle - \tilde{c}_1)\, A^1 \right] J^1.
\end{aligned}$$

□

The other cases

$$\frac{\partial \tilde{v}_{ki_1}}{\partial v_{k,i_1}}, \quad \frac{\partial \tilde{v}_{ki}}{\partial v_{k,i_1}}, \quad \frac{\partial \tilde{x}_{k,i_1}}{\partial v_{k,i_1}}, \quad \frac{\partial \tilde{x}_{k,i}}{\partial v_{k,i_1}}$$

are symmetric. The lemmas above introduced new expressions that require explanation. We provide them only for particle i, since particle i_1 has a symmetric form, and we remark that, at least for the expression below, they never mix. Recall

$$A := \frac{\partial \tilde{n}}{\partial \tilde{\theta}} \frac{\partial \tilde{\theta}}{\partial v'_{k,i}}.$$

The first part is straightforward

Lemma 2.6. *Let*

$$\tilde{n} = \frac{f(\tilde{\theta})}{\|f(\tilde{\theta})\|},$$

where

$$f(\tilde{\theta}) = \big(r_{\tilde{\theta}} \sin(2\pi\tilde{\theta}) + 2\pi r \cos(2\pi\tilde{\theta}),\ -r_{\tilde{\theta}} \cos(2\pi\tilde{\theta}) + 2\pi r \sin(2\pi\tilde{\theta})\big).$$

Then

$$\frac{\partial \tilde{n}}{\partial \tilde{\theta}} = \frac{f'(\tilde{\theta}) - \dfrac{f(\tilde{\theta}) \cdot f'(\tilde{\theta})}{\|f(\tilde{\theta})\|^2} f(\tilde{\theta})}{\|f(\tilde{\theta})\|} = \frac{1}{||f(\tilde{\theta})||} \left(I - \tilde{n}\tilde{n}^T\right) \frac{\partial f}{\partial \tilde{\theta}}$$

Proof. Looking at the equation for $\tilde{n}$ in (7), we see that is of the form

$$\tilde{n} = \frac{f(\tilde{\theta})}{||f(\tilde{\theta})||}.$$

The derivative can therefore be computed as

$$\frac{\partial \tilde{n}}{\partial \tilde{\theta}} = \frac{f'(\tilde{\theta}) - \frac{f(\tilde{\theta}) \cdot f'(\tilde{\theta})}{\|f(\tilde{\theta})\|^2} f(\tilde{\theta})}{\|f(\tilde{\theta})\|}$$

where $f(\tilde{\theta}) = (r_{\tilde{\theta}} \sin(2\pi\tilde{\theta}) + 2\pi r \cos(2\pi\tilde{\theta}), -r_{\tilde{\theta}} \cos(2\pi\tilde{\theta}) + 2\pi r \sin(2\pi\tilde{\theta}))$. This computation is straightforward, but it's lengthy, so I'll leave it for now. □

The second part uses (9)

Lemma 2.7.

$$\frac{\partial \tilde{\theta}}{\partial v'_{k,i}} = \frac{r(\tilde{\theta}) \begin{bmatrix} \sin 2\pi\tilde{\theta} \\ -\cos 2\pi\tilde{\theta} \end{bmatrix} + \begin{bmatrix} -x^y_{k,i} \\ x^x_{k,i} \end{bmatrix}}{r_{\tilde{\theta}}\big(v^y_{k,i} \cos 2\pi\tilde{\theta} - v^x_{k,i} \sin 2\pi\tilde{\theta}\big) - 2\pi r(\tilde{\theta})\big(v^y_{k,i} \sin 2\pi\tilde{\theta} + v^x_{k,i} \cos 2\pi\tilde{\theta}\big)}.$$

Proof. This term can be implicitly computed by recalling (9). We have

$$(v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta}))r(\tilde{\theta}, C) = v^y_{k,i}x^x_{k,i} - v^x_{k,i}x^y_{k,i} \tag{19}$$

Taking the derivative with respect to $v^x_{k,i}$ we get

$$\left(-2\pi v^y_{k,i}\sin(2\pi\tilde{\theta})\tfrac{\partial\tilde{\theta}}{\partial v^x_{k,i}} - \sin(2\pi\tilde{\theta}) - 2\pi v^x_{k,i}\cos(2\pi\tilde{\theta})\tfrac{\partial\tilde{\theta}}{\partial v^x_{k,i}}\right) r(\tilde{\theta}) + (v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta}))r_{\tilde{\theta}}(\tilde{\theta})\tfrac{\partial\tilde{\theta}}{\partial v^x_{k,i}} = -x^y_{k,i}$$

factoring out $\frac{\partial\tilde{\theta}}{\partial v^x_{k,i}}$ we get

$$\frac{\partial\tilde{\theta}}{\partial v^x_{k,i}} = \frac{-x^y_{k,i} + r(\tilde{\theta})\sin(2\pi\tilde{\theta})}{r_{\tilde{\theta}}(\tilde{\theta})(v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta})) - 2\pi r(\tilde{\theta})\left(v^x_{k,i}\cos(2\pi\tilde{\theta}) + v^y_{k,i}\sin(2\pi\tilde{\theta})\right)}.$$

The expression for $\frac{\partial\tilde{\theta}}{\partial v^y_{k,i}}$ is similar

$$\frac{\partial\tilde{\theta}}{\partial v^y_{k,i}} = \frac{x^x_{k,i} - r(\tilde{\theta})\cos(2\pi\tilde{\theta})}{r_{\tilde{\theta}}(\tilde{\theta})(v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta})) - 2\pi r(\tilde{\theta})\left(v^x_{k,i}\cos(2\pi\tilde{\theta}) + v^y_{k,i}\sin(2\pi\tilde{\theta})\right)}.$$

Combining we get

$$\frac{\partial\tilde{\theta}}{\partial v'_{k,i}} = \frac{r(\tilde{\theta})\begin{bmatrix}\sin 2\pi\tilde{\theta}\\ -\cos 2\pi\tilde{\theta}\end{bmatrix} + \begin{bmatrix}-x^y_{k,i}\\ x^x_{k,i}\end{bmatrix}}{r_{\tilde{\theta}}\big(v^y_{k,i}\cos 2\pi\tilde{\theta} - v^x_{k,i}\sin 2\pi\tilde{\theta}\big) - 2\pi r(\tilde{\theta})\big(v^y_{k,i}\sin 2\pi\tilde{\theta} + v^x_{k,i}\cos 2\pi\tilde{\theta}\big)}.$$

□

The last one is

Lemma 2.8.

$$\frac{\partial\tilde{c}}{\partial v'_{k,i}} = \langle\tilde{n}, \begin{bmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{bmatrix}\rangle\frac{\partial\tilde{r}}{\partial\tilde{\theta}}\frac{\partial\tilde{\theta}}{\partial v'_{k,i}} + \tilde{r}\left([\cos(2\pi\tilde{\theta}), \sin(2\pi\tilde{\theta})]\frac{\partial\tilde{n}}{\partial\tilde{\theta}}\frac{\partial\tilde{\theta}}{\partial v'_{k,i}} + 2\pi\tilde{n}^\top\begin{bmatrix}-\sin(2\pi\tilde{\theta})\\ \cos(2\pi\tilde{\theta})\end{bmatrix}\frac{\partial\tilde{\theta}}{\partial v'_{k,i}}\right) \tag{20}$$

Proof. Recall equation (11)

$$\tilde{c} = \tilde{r}\langle\tilde{n}, \begin{bmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{bmatrix}\rangle.$$

Differentiating as a product, we get the formula. □

2.2.2 Computing $\frac{\partial(\mathcal{J}_2+\mathcal{J}_3)}{\partial x_{k,i}}$

We now compute the gradient of the Lagrangian with respect to $x_{k,i}$ and $x'_{k,i}$. Since the objective function and $\mathcal{J}_2$ do not depend on either of the two state variables, we focus on $\mathcal{J}_3$

$$\frac{\partial \mathcal{J}_3}{\partial x_{k,i}} = \frac{1}{N}\left(\alpha_{k,i} - \left(\frac{\partial \tilde{x}_{k,i}}{\partial x_{k,i}}\right)^\top \alpha_{k+1,i}\right) \tag{21}$$

and similarly

$$\frac{\partial \mathcal{J}_3}{\partial x_{k,i_1}} = \frac{1}{N}\left(\alpha_{k,i_1} - \left(\frac{\partial \tilde{x}_{k,i_1}}{\partial x_{k,i_1}}\right)^\top \alpha_{k+1,i_1}\right). \tag{22}$$

The new expression is

$$\frac{\partial \tilde{x}_{k,i}}{\partial x_{k,i}}, \quad \frac{\partial \tilde{x}_{k,i_1}}{\partial x_{k,i_1}},$$

We will show one and remark that the other is similar due to symmetry.

Lemma 2.9.

$$\frac{\partial \tilde{x}_{k,i}}{\partial x_{k,i}} = \begin{cases} I & \textit{if } x'_{k,i} \in \Omega \\ I - 2\tilde{n}\,\tilde{n}^\top - 2\left(\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right)\frac{\partial \tilde{n}}{\partial \tilde{\theta}} + \tilde{n}\left((x'_{k,i})^\top \frac{\partial \tilde{n}}{\partial \tilde{\theta}} - F\right)\right)\frac{\partial \tilde{\theta}}{\partial x_{k,i}}. & \textit{Otherwise} \end{cases}$$

where

$$F = \left(\langle \tilde{n}, \begin{bmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{bmatrix}\rangle \frac{\partial \tilde{r}}{\partial \tilde{\theta}} + \tilde{r}\left(\begin{bmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{bmatrix}^\top \frac{\partial \tilde{n}}{\partial \tilde{\theta}} + 2\pi\tilde{n}^\top \begin{bmatrix}-\sin(2\pi\tilde{\theta})\\ \cos(2\pi\tilde{\theta})\end{bmatrix}\right)\right) = \frac{\partial \tilde{c}}{\partial \tilde{\theta}}.$$

Proof. We apply the chain rule

$$\frac{\partial}{\partial x_{k,i}}\left(2\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right)\tilde{n}\right) = 2\left(\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right)\frac{\partial \tilde{n}}{\partial \tilde{\theta}} + \tilde{n}\left((x'_{k,i})^\top \frac{\partial \tilde{n}}{\partial \tilde{\theta}} - F\right)\right)\frac{\partial \tilde{\theta}}{\partial x_{k,i}} + 2\tilde{n}\,\tilde{n}^\top.$$

□

The term

$$\frac{\partial \tilde{\theta}}{\partial x_{k,i}}$$

is computed the following way

Lemma 2.10.

$$\frac{\partial \tilde{\theta}}{\partial x_{k,i}} = \frac{\begin{bmatrix} v^y_{k,i} \\ -v^x_{k,i}\end{bmatrix}}{r_{\tilde{\theta}}\left(v^y_{k,i}\cos 2\pi\tilde{\theta} - v^x_{k,i}\sin 2\pi\tilde{\theta}\right) - 2\pi r(\tilde{\theta})\left(v^y_{k,i}\sin 2\pi\tilde{\theta} + v^x_{k,i}\cos 2\pi\tilde{\theta}\right)}.$$

Proof. Differentiating (19) with respect to $x^x_{k,i}$ and $x^y_{k,i}$ we get

$$\frac{\partial\tilde{\theta}}{\partial x^x_{k,i}} = \frac{v^y_{k,i}}{r_{\tilde{\theta}}(\tilde{\theta})(v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta})) - 2\pi r(\tilde{\theta})\left(v^x_{k,i}\cos(2\pi\tilde{\theta}) + v^y_{k,i}\sin(2\pi\tilde{\theta})\right)}.$$

and

$$\frac{\partial\tilde{\theta}}{\partial x^y_{k,i}} = \frac{-v^x_{k,i}}{r_{\tilde{\theta}}(\tilde{\theta})(v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta})) - 2\pi r(\tilde{\theta})\left(v^x_{k,i}\cos(2\pi\tilde{\theta}) + v^y_{k,i}\sin(2\pi\tilde{\theta})\right)}.$$

respectively. □

With all the above terms computed, we equate the derivatives of state variables to 0 and get the following rules for the evolution of the adjoint variables β and α.

Proposition 2.1. *If both $x'_{k,i}$, $x'_{k,i_1} \in \Omega$, then*

$$\begin{pmatrix}\beta_{k,i}\\ \beta_{k,i_1}\end{pmatrix} = \left[\frac{\partial(v'_{k,i}, v'_{k,i_1})}{\partial(v_{k,i}, v_{k,i_1})}\right]^\top \begin{pmatrix}\beta_{k+1,i} + \Delta t\alpha_{k+1,i}\\ \beta_{k+1,i_1} + \Delta t\alpha_{k+1,i_1}\end{pmatrix}.$$

If neither $x'_{k,i}$, $x'_{k,i_1} \notin \Omega$, then

$$\begin{pmatrix}\beta_{k,i}\\ \beta_{k,i_1}\end{pmatrix} = \left[\frac{\partial(v'_{k,i}, v'_{k,i_1})}{\partial(v_{k,i}, v_{k,i_1})}\right]^\top \begin{pmatrix}M_{k,i}\,\beta_{k+1,i} + N_{k,i}\,\alpha_{k+1,i}\\ M^1_{k,i_1}\,\beta_{k+1,i_1} + N^1_{k,i_1}\,\alpha_{k+1,i_1}\end{pmatrix} \tag{23}$$

Where

$$\begin{aligned}
M_{k,i} &= I - 2\langle\tilde{n}, v'_{k,i}\rangle A - 2\tilde{n}\left((v'_{k,i})^\top A + \tilde{n}^\top\right),\\
M^1_{k,i_1} &= I - 2\langle\tilde{n}_1, v'_{k,i_1}\rangle A^1 - 2\tilde{n}_1\left((v'_{k,i_1})^\top A^1 + \tilde{n}_1^\top\right),\\
N_{k,i} &= \Delta t\, I - 2\tilde{n}\left((x'_{k,i})^\top A + \Delta t\,\tilde{n}^\top - \frac{\partial\tilde{c}}{\partial v'_{k,i}}\right) - 2\left(\langle\tilde{n}, x'_{k,i}\rangle - \tilde{c}\right)A,\\
N^1_{k,i_1} &= \Delta t\, I - 2\tilde{n}_1\left((x'_{k,i_1})^\top A^1 + \Delta t\,\tilde{n}_1^\top - \frac{\partial\tilde{c}_1}{\partial v'_{k,i_1}}\right) - 2\left(\langle\tilde{n}_1, x'_{k,i_1}\rangle - \tilde{c}_1\right)A^1.
\end{aligned} \tag{24}$$

If $x'_{k,i} \in \Omega$ but $x'_{k,i_1} \notin \Omega$, then

$$\begin{pmatrix}\beta_{k,i}\\ \beta_{k,i_1}\end{pmatrix} = \left[\frac{\partial(v'_{k,i}, v'_{k,i_1})}{\partial(v_{k,i}, v_{k,i_1})}\right]^\top \begin{pmatrix}\beta_{k+1,i} + \Delta t\,\alpha_{k+1,i}\\ M^1_{k,i_1}\,\beta_{k+1,i_1} + N^1_{k,i_1}\,\alpha_{k+1,i_1}.\end{pmatrix} \tag{25}$$

The last case is symmetric

$$\begin{pmatrix}\beta_{k,i}\\ \beta_{k,i_1}\end{pmatrix} = \left[\frac{\partial(v'_{k,i}, v'_{k,i_1})}{\partial(v_{k,i}, v_{k,i_1})}\right]^\top \begin{pmatrix}M_{k,i}\,\beta_{k+1,i} + N_{k,i}\,\alpha_{k+1,i}\\ \beta_{k+1,i_1} + \Delta t\,\alpha_{k+1,i_1}\end{pmatrix} \tag{26}$$

Proof. We begin with the case where $x'_{k,i}, x'_{k,i_1} \in \Omega$. In case equations (17) and (18) become

$$\beta_{k,i} = \left(\frac{\partial v'_{k,i}}{\partial v_{k,i}}\right)^\top \beta_{k+1,i} + \left(\frac{\partial v'_{k,i_1}}{\partial v_{k,i}}\right)^\top \beta_{k+1,i_1} + \Delta t \left(\frac{\partial v'_{k,i}}{\partial v_{k,i}}\right)^\top \alpha_{k+1,i} + \Delta t \left(\frac{\partial v'_{k,i_1}}{\partial v_{k,i}}\right)^\top \alpha_{k+1,i_1} \tag{27}$$

and

$$\beta_{k,i_1} = \left(\frac{\partial v'_{k,i_1}}{\partial v_{k,i_1}}\right)^\top \beta_{k+1,i_1} + \left(\frac{\partial v'_{k,i}}{\partial v_{k,i_1}}\right)^\top \beta_{k+1,i} + \Delta t \left(\frac{\partial v'_{k,i_1}}{\partial v_{k,i_1}}\right)^\top \alpha_{k+1,i} + \Delta t \left(\frac{\partial v'_{k,i}}{\partial v_{k,i_1}}\right)^\top \alpha_{k+1,i_1}. \tag{28}$$

This can be written in a more compact way as

$$\begin{pmatrix} \beta_{k,i} \\ \beta_{k,i_1} \end{pmatrix} = \left[\frac{\partial(v'_{k,i}, v'_{k,i_1})}{\partial(v_{k,i}, v_{k,i_1})}\right]^\top \begin{pmatrix} \beta_{k+1,i} + \Delta t \alpha_{k+1,i} \\ \beta_{k+1,i_1} + \Delta t \alpha_{k+1,i_1} \end{pmatrix}.$$

Now we look at the case where both $x'_{k,i}, x'_{k,i_1} \notin \Omega$. Then the above equations become

$$\begin{aligned}
\beta_{k,i} = &\left(\left[I - 2\langle \tilde{n}, v'_{k,i}\rangle A - 2\tilde{n}\left((v'_{k,i})^\top A + \tilde{n}^\top\right)\right] \frac{\partial v'_{k,i}}{\partial v_{k,i}}\right)^\top \beta_{k+1,i} \\
&+ \left(\left[I - 2\langle \tilde{n}_1, v'_{k,i_1}\rangle A^1 - 2\tilde{n}_1\left((v'_{k,i_1})^\top A^1 + \tilde{n}_1^\top\right)\right] J^1\right)^\top \beta_{k+1,i_1} \\
&+ \left(\left[\Delta t I - 2\tilde{n}\left((x'_{k,i})^\top A + \Delta t \tilde{n}^\top - \frac{\partial \tilde{c}}{\partial v'_{k,i}}\right) - 2\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right) A\right] \frac{\partial v'_{k,i}}{\partial v_{k,i}}\right)^\top \alpha_{k+1,i} \\
&+ \left(\left[\Delta t I - 2\tilde{n}_1\left((x'_{k,i_1})^\top A^1 + \Delta t\, \tilde{n}_1^\top - \frac{\partial \tilde{c}_1}{\partial v'_{k,i_1}}\right) - 2(\langle \tilde{n}_1, x'_{k,i_1}\rangle - \tilde{c}_1)\, A^1\right] J^1\right)^\top \alpha_{k+1,i_1}
\end{aligned} \tag{29}$$

and similarly

$$\begin{aligned}
\beta_{k,i_1} = &\left(\left[I - 2\langle \tilde{n}_1, v'_{k,i_1}\rangle A^1 - 2\tilde{n}_1\left((v'_{k,i_1})^\top A^1 + \tilde{n}_1^\top\right)\right] \frac{\partial v'_{k,i_1}}{\partial v_{k,i_1}}\right)^\top \beta_{k+1,i_1} \\
&+ \left(\left[I - 2\langle \tilde{n}, v'_{k,i}\rangle A - 2\tilde{n}\left((v'_{k,i})^\top A + \tilde{n}^\top\right)\right] \frac{\partial v'_{k,i}}{\partial v_{k,i_1}}\right)^\top \beta_{k+1,i} \\
&+ \left(\left[\Delta t I - 2\tilde{n}_1\left((x'_{k,i_1})^\top A^1 + \Delta t\, \tilde{n}_1^\top - \frac{\partial \tilde{c}_1}{\partial v'_{k,i_1}}\right) - 2\left(\langle \tilde{n}_1, x'_{k,i_1}\rangle - \tilde{c}_1\right) A^1\right] \frac{\partial v'_{k,i_1}}{\partial v_{k,i_1}}\right)^\top \alpha_{k+1,i_1} \\
&+ \left(\left[\Delta t I - 2\tilde{n}\left((x'_{k,i})^\top A + \Delta t\, \tilde{n}^\top - \frac{\partial \tilde{c}}{\partial v'_{k,i}}\right) - 2(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c})\, A\right] \frac{\partial v'_{k,i}}{\partial v_{k,i_1}}\right)^\top \alpha_{k+1,i}
\end{aligned} \tag{30}$$

Combining them, we get

$$\begin{pmatrix} \beta_{k,i} \\ \beta_{k,i_1} \end{pmatrix} = \left[\frac{\partial(v'_{k,i}, v'_{k,i_1})}{\partial(v_{k,i}, v_{k,i_1})} \right]^\top \begin{pmatrix} M_{k,i}\, \beta_{k+1,i} + N_{k,i}\, \alpha_{k+1,i} \\ M^1_{k,i_1}\, \beta_{k+1,i_1} + N^1_{k,i_1}\, \alpha_{k+1,i_1} \end{pmatrix} \tag{31}$$

Where

$$\begin{aligned} M_{k,i} &= I - 2\langle \tilde{n}, v'_{k,i} \rangle A - 2\tilde{n} \left((v'_{k,i})^\top A + \tilde{n}^\top \right), \\ M^1_{k,i_1} &= I - 2\langle \tilde{n}_1, v'_{k,i_1} \rangle A^1 - 2\tilde{n}_1 \left((v'_{k,i_1})^\top A^1 + \tilde{n}_1^\top \right), \\ N_{k,i} &= \Delta t\, I - 2\tilde{n} \left((x'_{k,i})^\top A + \Delta t\, \tilde{n}^\top - \frac{\partial \tilde{c}}{\partial v'_{k,i}} \right) - 2 \left(\langle \tilde{n}, x'_{k,i} \rangle - \tilde{c} \right) A, \\ N^1_{k,i_1} &= \Delta t\, I - 2\tilde{n}_1 \left((x'_{k,i_1})^\top A^1 + \Delta t\, \tilde{n}_1^\top - \frac{\partial \tilde{c}_1}{\partial v'_{k,i_1}} \right) - 2 \left(\langle \tilde{n}_1, x'_{k,i_1} \rangle - \tilde{c}_1 \right) A^1. \end{aligned} \tag{32}$$

The mixed case follows from taking a suitable combination of both. □

Proposition 2.2. *If both $x'_{k,i}$, $x'_{k,i_1} \in \Omega$, then*

$$\begin{pmatrix} \alpha_{k,i} \\ \alpha_{k,i_1} \end{pmatrix} = \begin{pmatrix} \alpha_{k+1,i} \\ \alpha_{k+1,i_1} \end{pmatrix}. \tag{33}$$

If neither $x'_{k,i}$, $x'_{k,i_1} \notin \Omega$, then

$$\begin{pmatrix} \alpha_{k,i} \\ \alpha_{k,i_1} \end{pmatrix} = \begin{pmatrix} G_{k,i}^\top\, \alpha_{k+1,i} \\ {G^1_{k,i_1}}^\top\, \alpha_{k+1,i_1} \end{pmatrix} \tag{34}$$

where

$$\begin{aligned} G_{k,i} &= I - 2 \left(\left(\langle \tilde{n}, x'_{k,i} \rangle - \tilde{c} \right) \frac{\partial \tilde{n}}{\partial \tilde{\theta}} + \tilde{n} \left((x'_{k,i})^\top \frac{\partial \tilde{n}}{\partial \tilde{\theta}} - F \right) \right) \frac{\partial \tilde{\theta}}{\partial x_{k,i}} - 2\tilde{n}\, \tilde{n}^\top, \\ G^1_{k,i_1} &= I - 2 \left(\left(\langle \tilde{n}_1, x'_{k,i_1} \rangle - \tilde{c}_1 \right) \frac{\partial \tilde{n}_1}{\partial \tilde{\theta}_1} + \tilde{n}_1 \left((x'_{k,i_1})^\top \frac{\partial \tilde{n}_1}{\partial \tilde{\theta}_1} - F_1 \right) \right) \frac{\partial \tilde{\theta}_1}{\partial x_{k,i_1}} - 2\tilde{n}_1\, \tilde{n}_1^\top. \end{aligned} \tag{35}$$

If $x'_{k,i} \in \Omega$ but $x'_{k,i_1} \notin \Omega$, then

$$\begin{pmatrix} \alpha_{k,i} \\ \alpha_{k,i_1} \end{pmatrix} = \begin{pmatrix} \alpha_{k+1,i} \\ {G^1_{k,i_1}}^\top\, \alpha_{k+1,i_1} \end{pmatrix} \tag{36}$$

The last case is symmetric

$$\begin{pmatrix} \alpha_{k,i} \\ \alpha_{k,i_1} \end{pmatrix} = \begin{pmatrix} G_{k,i}^\top\, \alpha_{k+1,i} \\ \alpha_{k+1,i_1} \end{pmatrix} \tag{37}$$

Proof. We begin with the case where $x'_{k,i}, x'_{k,i_1} \in \Omega$. In case equations (21) and (22) become

$$\begin{pmatrix} \alpha_{k,i} \\ \alpha_{k,i_1} \end{pmatrix} = \begin{pmatrix} \alpha_{k+1,i} \\ \alpha_{k+1,i_1} \end{pmatrix}.$$

If neither $x'_{k,i}$, $x'_{k,i_1} \notin \Omega$, then

$$\begin{pmatrix} \alpha_{k,i} \\ \alpha_{k,i_1} \end{pmatrix} = \begin{pmatrix} \left[2\left(\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right)\frac{\partial \tilde{n}}{\partial \tilde{\theta}} + \tilde{n}\left((x'_{k,i})^\top \frac{\partial \tilde{n}}{\partial \tilde{\theta}} - F\right)\right)\frac{\partial \tilde{\theta}}{\partial x_{k,i}} + 2\tilde{n}\,\tilde{n}^\top\right]^\top \alpha_{k+1,i} \\ \left[2\left(\left(\langle \tilde{n}_1, x'_{k,i_1}\rangle - \tilde{c}_1\right)\frac{\partial \tilde{n}_1}{\partial \tilde{\theta}_1} + \tilde{n}_1\left((x'_{k,i_1})^\top \frac{\partial \tilde{n}_1}{\partial \tilde{\theta}_1} - F_1\right)\right)\frac{\partial \tilde{\theta}_1}{\partial x_{k,i_1}} + 2\tilde{n}_1\,\tilde{n}_1^\top\right]^\top \alpha_{k+1,i_1} \end{pmatrix}.$$

The last two cases are a combination of both. □

3 Computing $\nabla_{\vec{C}}\mathcal{J}$

The goal of this section is to compute the gradient $\nabla_{\vec{C}}\mathcal{J}$.

$$\frac{\partial \mathcal{J}}{\partial \vec{C}} = -\frac{1}{N}\sum_{k=0}^{M-1}\sum_{i=1}^{N}\left(\frac{\partial \tilde{v}_{k,i}}{\partial \vec{C}}\right)^\top \beta_{k+1,i} - \frac{1}{N}\sum_{k=0}^{M-1}\sum_{i=1}^{N}\left(\frac{\partial \tilde{x}_{k,i}}{\partial \vec{C}}\right)^\top \alpha_{k+1,i}$$

We compute each term separately.

Lemma 3.1.

$$\frac{\partial \tilde{v}_{k,i}}{\partial \vec{C}} = \begin{cases} 0 & \textit{if } x'_{k,i} \in \Omega \\ -2\left[\langle \tilde{n}, v'_{k,i}\rangle I + \tilde{n}(v'_{k,i})^\top\right]\frac{\partial \tilde{n}}{\partial \vec{C}} & \textit{Otherwise} \end{cases}$$

Proof. If $x'_{k,i} \in \Omega$, then $\tilde{v}_{k,i} = v'_{k,i}$ which is independent of $\vec{C}$ so the gradient inside of the domain equals 0. If $x'_{k,i} \notin \Omega$, then

$$\frac{\partial}{\partial \vec{C}}\left(v'_{k,i} - 2\langle \tilde{n}, v'_{k,i}\rangle \tilde{n}\right) = -2\left[\langle \tilde{n}, v'_{k,i}\rangle I + \tilde{n}(v'_{k,i})^\top\right]\frac{\partial \tilde{n}}{\partial \vec{C}}.$$

□

Lemma 3.2.

$$\frac{\partial \tilde{x}_{k,i}}{\partial \vec{C}} = \begin{cases} 0 & \textit{If } x'_{k,i} \in \Omega \\ -2\left[\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right) I + \tilde{n}(x'_{k,i})^\top\right]\frac{\partial \tilde{n}}{\partial \vec{C}} + 2\tilde{n}\frac{\partial \tilde{c}}{\partial \vec{C}} & \textit{Otherwise} \end{cases}$$

Proof. If $x'_{k,i} \in \Omega$, then $\tilde{x}_{k,i} = x'_{k,i}$ which is independent of $\vec{C}$ so the gradient inside of the domain equals 0. If $x'_{k,i} \notin \Omega$, then

$$\frac{\partial}{\partial \vec{C}}\left(x'_{k,i} - 2\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right)\tilde{n}\right) = -2\left[\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right) I + \tilde{n}(x'_{k,i})^\top\right]\frac{\partial \tilde{n}}{\partial \vec{C}} + 2\tilde{n}\frac{\partial \tilde{c}}{\partial \vec{C}}$$

□

This leaves us with computing

$$\frac{\partial \tilde{n}}{\partial \vec{C}}, \frac{\partial \tilde{c}}{\partial \vec{C}}$$

Lemma 3.3.

$$\frac{\partial \tilde{n}}{\partial \vec{C}} = \frac{1}{\|f\|}\left(I - \tilde{n}\tilde{n}^\top\right)\frac{\partial f}{\partial \vec{C}}.$$

where

$$\frac{\partial f}{\partial \vec{C}} = \left[\frac{dP}{d\tilde{\theta}}\begin{pmatrix} r_{\tilde{\theta}} \\ r \end{pmatrix} + P\begin{pmatrix} r_{\tilde{\theta}\tilde{\theta}} \\ r_{\tilde{\theta}} \end{pmatrix}\right]\frac{\partial \tilde{\theta}}{\partial \vec{C}} + P(\tilde{\theta})\begin{pmatrix} \frac{\partial r_{\tilde{\theta}}}{\partial \vec{C}} \\ \frac{\partial r}{\partial \vec{C}} \end{pmatrix},$$

$$\frac{\partial \tilde{\theta}}{\partial \vec{C}} = -\frac{v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta})}{r_{\tilde{\theta}}\left[v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta})\right] - 2\pi r\left[v^y_{k,i}\sin(2\pi\tilde{\theta}) + v^x_{k,i}\cos(2\pi\tilde{\theta})\right]}\frac{\partial r}{\partial \vec{C}},$$

$$\frac{\partial r}{\partial \vec{C}} = \begin{pmatrix} 1 \\ \cos(2\pi\tilde{\theta}) \\ \vdots \\ \cos(2\pi N\tilde{\theta}) \\ \sin(2\pi\tilde{\theta}) \\ \vdots \\ \sin(2\pi N\tilde{\theta}) \end{pmatrix}^\top, \qquad \frac{\partial r_{\tilde{\theta}}}{\partial \vec{C}} = \begin{pmatrix} 0 \\ -2\pi\sin(2\pi\tilde{\theta}) \\ \vdots \\ -2\pi N\sin(2\pi N\tilde{\theta}) \\ 2\pi\cos(2\pi\tilde{\theta}) \\ \vdots \\ 2\pi N\cos(2\pi N\tilde{\theta}) \end{pmatrix}^\top,$$

and

$$\frac{dP}{d\tilde{\theta}} = 2\pi\begin{pmatrix} \cos(2\pi\tilde{\theta}) & -2\pi\sin(2\pi\tilde{\theta}) \\ \sin(2\pi\tilde{\theta}) & 2\pi\cos(2\pi\tilde{\theta}) \end{pmatrix}.$$

Proof. As in the proof of Lemma 2.6,

$$\frac{\partial \tilde{n}}{\partial \vec{C}} = \frac{1}{||f||}\left(I - \tilde{n}\tilde{n}^T\right)\frac{\partial f}{\partial \vec{C}}$$

where $f(\tilde{\theta}) = P(\tilde{\theta})\begin{pmatrix} r_{\tilde{\theta}} \\ r \end{pmatrix}$ and

$$P(\tilde{\theta}) = \begin{pmatrix} \sin(2\pi\tilde{\theta}) & 2\pi\cos(2\pi\tilde{\theta}) \\ -\cos(2\pi\tilde{\theta}) & 2\pi\sin(2\pi\tilde{\theta}) \end{pmatrix}.$$

Applying the product rule

$$\frac{\partial f}{\partial \vec{C}} = \frac{\partial P}{\partial \vec{C}}\begin{pmatrix} r_{\tilde{\theta}} \\ r \end{pmatrix} + P(\tilde{\theta})\frac{\partial}{\partial \vec{C}}\begin{pmatrix} r_{\tilde{\theta}} \\ r \end{pmatrix}$$

Since P depends on $\vec{C}$ only through $\tilde{\theta}$ the first summand becomes

$$\frac{\partial P}{\partial \vec{C}} = \frac{dP}{d\tilde{\theta}} \frac{\partial \tilde{\theta}}{\partial \vec{C}}.$$

The second summand depends on $\vec{C}$ both through $\tilde{\theta}$ and r

$$\frac{\partial}{\partial \vec{C}} \begin{pmatrix} r_{\tilde{\theta}} \\ r \end{pmatrix} = \begin{pmatrix} r_{\tilde{\theta}\tilde{\theta}} \\ r_{\tilde{\theta}} \end{pmatrix} \frac{\partial \tilde{\theta}}{\partial \vec{C}} + \begin{pmatrix} \frac{\partial r_{\tilde{\theta}}}{\partial \vec{C}} \\ \frac{\partial r}{\partial \vec{C}} \end{pmatrix}.$$

Combining the terms gets us

$$\frac{\partial f}{\partial \vec{C}} = \left[\frac{dP}{d\tilde{\theta}} \begin{pmatrix} r_{\tilde{\theta}} \\ r \end{pmatrix} + P \begin{pmatrix} r_{\tilde{\theta}\tilde{\theta}} \\ r_{\tilde{\theta}} \end{pmatrix} \right] \frac{\partial \tilde{\theta}}{\partial \vec{C}} + P(\tilde{\theta}) \begin{pmatrix} \frac{\partial r_{\tilde{\theta}}}{\partial \vec{C}} \\ \frac{\partial r}{\partial \vec{C}} \end{pmatrix}.$$

From the definitions we have

$$\frac{\partial r}{\partial \vec{C}} = \begin{pmatrix} 1 \\ \cos(2\pi\tilde{\theta}) \\ \vdots \\ \cos(2\pi N\tilde{\theta}) \\ \sin(2\pi\tilde{\theta}) \\ \vdots \\ \sin(2\pi N\tilde{\theta}) \end{pmatrix}^\top, \qquad \frac{\partial r_{\tilde{\theta}}}{\partial \vec{C}} = \begin{pmatrix} 0 \\ -2\pi\sin(2\pi\tilde{\theta}) \\ \vdots \\ -2\pi N\sin(2\pi N\tilde{\theta}) \\ 2\pi\cos(2\pi\tilde{\theta}) \\ \vdots \\ 2\pi N\cos(2\pi N\tilde{\theta}) \end{pmatrix}^\top$$

and

$$\frac{dP}{d\tilde{\theta}} = 2\pi \begin{pmatrix} \cos(2\pi\tilde{\theta}) & -2\pi\sin(2\pi\tilde{\theta}) \\ \sin(2\pi\tilde{\theta}) & 2\pi\cos(2\pi\tilde{\theta}) \end{pmatrix}.$$

That leaves us only with the term below left

$$\frac{\partial \tilde{\theta}}{\partial \vec{C}}.$$

Recalling (10)

$$\left[v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta}) \right] r(\tilde{\theta}, \vec{C}) = v^y_{k,i}x^x_{k,i} - v^x_{k,i}x^y_{k,i}.$$

Differentiating with respect to $\vec{C}$ we get

$$\left[-2\pi v^y_{k,i}\sin(2\pi\tilde{\theta}) - 2\pi v^x_{k,i}\cos(2\pi\tilde{\theta}) \right] r \frac{\partial \tilde{\theta}}{\partial \vec{C}} + \left[v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta}) \right] \left(\frac{\partial r}{\partial \vec{C}} + \frac{\partial r}{\partial \tilde{\theta}} \frac{\partial \tilde{\theta}}{\partial \vec{C}} \right) = 0$$

or equivalently

$$\frac{\partial \tilde{\theta}}{\partial \vec{C}} = -\frac{v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta})}{r_{\tilde{\theta}}\left[v^y_{k,i}\cos(2\pi\tilde{\theta}) - v^x_{k,i}\sin(2\pi\tilde{\theta})\right] - 2\pi r\left[v^y_{k,i}\sin(2\pi\tilde{\theta}) + v^x_{k,i}\cos(2\pi\tilde{\theta})\right]}\frac{\partial r}{\partial \vec{C}}$$

□

There is one last expression we need to be able to compute

Lemma 3.4.

$$\frac{\partial \tilde{c}}{\partial \vec{C}} = \langle \tilde{n}, \begin{pmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{pmatrix}\rangle \frac{\partial \tilde{r}}{\partial \vec{C}} + \tilde{r}\left[\begin{pmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{pmatrix}^\top \frac{\partial \tilde{n}}{\partial \vec{C}} + 2\pi\tilde{n}^\top \begin{pmatrix}-\sin(2\pi\tilde{\theta})\\ \cos(2\pi\tilde{\theta})\end{pmatrix}\frac{\partial \tilde{\theta}}{\partial \vec{C}}\right]$$

Proof. Differentiating the equation

$$\tilde{c} = \tilde{r}\langle \tilde{n}, \begin{bmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{bmatrix}\rangle$$

with respect to $\vec{C}$ we get

$$\begin{aligned}\frac{\partial \tilde{c}}{\partial \vec{C}} &= \frac{\partial \tilde{r}}{\partial \vec{C}}\langle \tilde{n}, \begin{bmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{bmatrix}\rangle + \tilde{r}\frac{\partial}{\partial \vec{C}}\left(\langle \tilde{n}, \begin{bmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{bmatrix}\rangle\right)\\ &= \langle \tilde{n}, \begin{pmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{pmatrix}\rangle \frac{\partial \tilde{r}}{\partial \vec{C}} + \tilde{r}\left[\begin{pmatrix}\cos(2\pi\tilde{\theta})\\ \sin(2\pi\tilde{\theta})\end{pmatrix}^\top \frac{\partial \tilde{n}}{\partial \vec{C}} + 2\pi\tilde{n}^\top \begin{pmatrix}-\sin(2\pi\tilde{\theta})\\ \cos(2\pi\tilde{\theta})\end{pmatrix}\frac{\partial \tilde{\theta}}{\partial \vec{C}}\right]\end{aligned}$$

□

Putting it together, we see that the gradient is

$$\begin{aligned}\frac{\partial \mathcal{J}}{\partial \vec{C}} = &-\frac{1}{N}\sum_{k=0}^{M-1}\sum_{i\in\mathcal{I}_k}\left(-2\left[\langle \tilde{n}, v'_{k,i}\rangle I + \tilde{n}(v'_{k,i})^\top\right]\frac{\partial \tilde{n}}{\partial \vec{C}}\right)^\top \beta_{k+1,i}\\ &-\frac{1}{N}\sum_{k=0}^{M-1}\sum_{i\in\mathcal{I}_k}\left(-2\left[\left(\langle \tilde{n}, x'_{k,i}\rangle - \tilde{c}\right) I + \tilde{n}(x'_{k,i})^\top\right]\frac{\partial \tilde{n}}{\partial \vec{C}} + 2\tilde{n}\frac{\partial \tilde{c}}{\partial \vec{C}}\right)^\top \alpha_{k+1,i}\end{aligned}$$

Where $\mathcal{I}_k = \{i \in \{1,\dots,N\} : x'_{k,i} \notin \Omega\}$

References

[1] Michel C Delfour and J-P Zolésio. *Shapes and geometries: metrics, analysis, differential calculus, and optimization.* SIAM, 2011.

[2] Mo Zhou, Jiequn Han, Manas Rachh, and Carlos Borges. A neural network warm-start approach for the inverse acoustic obstacle scattering problem. *Journal of Computational Physics*, 490:112341, 2023.